

Gender and corruption in South Asian SME's credit access

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ABSTRACT. This paper explores the corrupt behaviour of small and medium enterprise (SME) owners in South Asia, using World Bank Enterprise Surveys firm level data. In particular, we investigate whether corruption differs between genders and if this difference relates to credit constraints faced by firms. Controlling potential endogeneity and reverse causality of corruption and credit constraints via instrument variables, this study supports the “grease the wheel” concept. We find that women are not necessarily more averse to corruption than men in South Asia. In particular, when female owned SMEs pay bribes, they have on average a 71% lower probability of being credit constrained. However, when male owned SMEs pay bribes, they have only on average a 58% lower probability of being credit constrained than their counterparts who do not pay bribes. This indicates that attitudes towards bribery as unethical may even be more a question of culture rather than a matter of gender. Who greases the wheel? Both genders!

KEYWORDS: Women, Corruption, Bribes, SME, Credit access, South Asia.

JEL Classification: D73, E5, G21, L25

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Introduction

This study examines how corruption affects female and male owned businesses and exacerbates gender-based asymmetries in access credit in South Asia¹. While it is often argued that female business owners face gender-specific challenges and might have different behaviours based on different ethics and moral standards, it is crucial to seek empirical evidence at the microeconomic level. According to the conventional wisdom, men are more willing than women to behave unethically and women are significantly more likely than men to view certain questionable acts as unethical. This has been demonstrated to be the case in a wide range of institutional contexts, through both experimental and survey-based studies. Literature covering ethics abounds, with findings suggesting women are more likely to take stronger stances on ethical behaviour (David, Fisman, & Gatti, 2001; Kray & Haselhuhn, 2012) and carry on all the more generously when confronted with financial and economic decisions. However, societal/cultural norms, personal/individual and situational factors may directly impact gender differences in relation to ethical behaviour. Ethical behaviour requires a clear distinction between right and wrong; however, this distinction is not always well-defined, nor it is necessarily the same for every individual (Lewis, 1985). Views on corruption and extortion may vary according to the institutional settings. Hence, corruptibility in women may be influenced by more than biology and may not be a universal phenomenon.

Looking at gender differences in the corrupt behaviour of small and medium enterprises' (SME)² owners is important. If women entrepreneurs are less likely to engage in corruption

¹ According to the United Nations geographical region classification, South Asia comprises the countries of Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan and Sri Lanka. This study excluded Maldives due to data unavailability and also excluded Sri Lanka due to data not being up to date (Sri Lanka Enterprise Survey collected data in 2011).

² The Enterprise Surveys that were undertaken by the World Bank in 2014 categorised South Asian firms having up to 99 employees as SMEs and that is the definition used in this study. Firms with 5 to 19 employees are categorised in the small firm category and firms with 20 to 99 employees are in medium size category.

than men entrepreneurs, then measures and policies to fight corruption will reduce the advantages male-owned enterprises gain from these behaviours and thereby impact any gender gap. In addition, a business environment that is friendlier to women-owned SMEs would be more attractive to women and so would encourage more women to become entrepreneurs, with potential for increasing their incomes and welfare. Such an environment would also be more conducive to the development of the business sector, which is important for poverty alleviation and sustainable growth in developing economies.

The credit market for SMEs provides a fertile ground for corruption for several reasons. Tough conditions are often imposed on SMEs for accessing loans, and banks face potentially high costs in serving SMEs. This creates a situation where incentives may be used to gain access to finance and to buy compliance of a credit agent. Also, specific features and structures of the SME-bank relationship, such as high levels of informality and blurred lines between business and personal lives of SME owners, further amplify corruption risks. According to UNIDO (United Nations Industrial Development Organization, 2007), SMEs pay more than US\$1 billion (equivalent to 3% of the world's GDP) in bribes every year. Although payments of bribes are detrimental to all types of businesses, it poses a major problem for the development of SMEs, as they pay a much higher percentage of their annual revenues in bribes to public officials than do larger companies ("Corruption and small medium enterprises (SMEs)," 2013).

In South Asian economies, the most recurrent problem for SME owners is related to access to credit, and corruption ("Enterprise Surveys," 2014). South Asia is a patriarchal society, and so women members of society have limited or no entitlement to immovable and movable property and they are not part of the "old boy's network" of senior public, political and commercial figures. In contexts where bribery has become a prerequisite for accessing services, rights, and resources, women's relatively weaker access to and control of personal wealth

means they are more frequently denied access to these services. In addition, female SME owners' limited access to public officials and low income level diminishes their ability to pay bribes, further restricting their access to credit. However, according to our knowledge, neither research nor policy has given consideration to the impact of corruption on either women or men owned SMEs.

Based on firm level data for five South Asian economies (Afghanistan, Bangladesh, India, Nepal, and Pakistan) our results indicate that regardless of a firm owner's gender, the majority of SMEs in South Asia need to pay bribes to gain access to credit. Further, consistent with the "grease the wheels" concept, we conclude that corruption may be beneficial in countries where other aspects of governance are ineffective. Results also indicate that corruption has differing impacts between women and men in terms of accessing credit. In particular, when female owned SMEs pay bribes, they are on average 71% more likely to be less credit constrained. However, when male owned SMEs pay bribes they are on average only 58% less credit constrained than their counterparts who do not pay bribes.

This study contributes to the understanding of corruption's impact in several ways. Prior studies mainly use macro data (see Dollar, Raymond, & Roberta, 2001; Swamy, Stephen, Young, & Omar, 2001), or when it comes to micro level analysis, most studies focus on a single country (see Tuyen, Huong, Doan, & Hiep, 2016; Vu, Tran, Nguyen, & Lim, 2016), whereas our cross-country study is, to our knowledge, the only work providing an insight into gender, corruption and credit access. Secondly, the current research is significant in establishing whether there is a causal relationship between SME corruption levels and credit constraints. The knowledge of whether growth is hindered by credit constraints or by corruption is essential when formulating efficacious public policy. This study uses an instrumental variable approach

with a two-step probit model³ controlling for endogeneity bias. Firm-level data from the World Bank are provided in 1-year slices and accordingly only cross-sectional analyses are possible. Corruption may be endogenous⁴ to credit constraints in several ways, hence creating econometric problems when modelling access to credit. The relationship of corruption and credit constraints is subject to reverse causation, potentially omitted variables and unobservable factors that might have a positive or negative effect. Controlling for the endogeneity of corruption reduces the likelihood of overestimating/underestimating the true effect of firm corruption on credit constraints.

Section 2 of the paper reviews prior research. It is followed, in section 3, with a discussion of the data, variables, methods and procedures used in the empirical study. The results are presented in section 4 and conclusions in section 5

Literature review and hypothesis development

Corruption is a major economic concern in emerging and developing countries, because it influences growth and productivity (Mauro, 1995; Méon & Weill, 2010; Wei, 2000). Absence of a strong institutional environment and lack of transparency in the administrative and legal frameworks give fertile ground to development of rent seeking behaviours in such countries (Myint, 2000). In addition to the rise of an informal economy and the high social costs associated with corruption (Myint, 2000), its adverse consequences arrest SMEs' development. Theoretical explanations concerning the effect of corruption on firms' access to credit suggest two predominant issues. One strand of literature argues that corruption is likely to have

³ ivprobit – Probit model with continuous endogenous regressors. It is typically for this reason that generalised linear models, like probit or logit, are used to model binary dependent variables in applied research, and an approach that extends the probit model to account for endogeneity. See the probit techniques proposed by Rivers and Vuong (1988).

⁴ In a statistical model, a parameter or variable is said to be endogenous when there is a correlation between the parameter or variable and the error term (Wooldridge, 2002).

beneficial effects for developing countries' firms suffering from restrictive private monopolies and government practices (Voskanyan, 2000). Corruption allows firms to obtain external finance such as bank loans (Khawaja & Mian, 2005) by overcoming bureaucratic processes and unclear or complex regulations (Agrawal & Knoeber, 2001). Supporting this “grease the wheels” concept, Weill (2011) employs bank-level data from all over the world to show that corruption (Swamy et al., 2001) can enhance bank lending. Also, Chen, Liu and Su (2013) find evidence supporting the positive impact of corruption on access to bank credit in China. They observe a positive link between a proxy for the size of bribes the firm provides and access of external credit.

On the other hand, considering theoretical analyses and empirical evidence, some researchers show corruption “sands the wheels” of growth (Mauro, 1995; Reinikka & Svensson, 2004). For example, corruption in lending can contribute to a reduction in firms' bank debt due to increasing the cost of the loan for the borrower. In this case, a bribe amounts to a tax on borrowers and so constitutes an obstacle to credit (Fungáčová, Kochanova, & Weil, 2014). SMEs have no established lobby groups and connections compared to their larger counterparts, so they are particularly subject to heavy bribes and expropriations. Furthermore, unlike large businesses, SMEs are often credit constrained (Beck, 2007) and they may not have cash to pay bribes. Consequently, bribes reduce SME investment and impact on their long-term survival. Moreover, corruption favours a particular class of people and creates inequality in opportunities when accessing bank credit and external financing (Mo, 2001). In addition to shrinking opportunities through impeding profitability, disparity in circumstances prompts dissatisfaction and socio-political precariousness (Akongdit, 2013).

Although recent research using micro level data emphasises the relationship between corruption and access to finance, the findings are inconclusive (Fungáčová et al., 2014; Kato

& Sato, 2014). This may be due to a causal relationship between corruption and access to finance in SMEs. However, it is not yet established whether the relationship is causal, and if it is, what the driving factor might be? Does a lack of access to formal finance encourage entrepreneurs to pay bribes to get access to the formal finance or does corruption prevent them from accessing formal finance? While keeping in mind the causality effect, this study hypothesises:

H₁: Paying bribes is positively related to SMEs' level of credit constraints in South Asia.

There is a growing body of research finding systematic differences in behavioural characteristics across gender (see Rivas, 2013). In the economic literature, over a decade ago a gender dimension was added through two classic papers by Swamy et al. (2001) and Dollar et al. (2001), both drawing on the thought that women act uniquely in contrast to men in numerous economic and financial circumstances. Both papers find that women in positions of influence are less frequently involved in corruption activities. In their cross-country study, analysing over 100 countries, Dollar et al. (2001) find a strong negative relationship between the level of female participation in politics and upper chambers and a corruption index. Similar to that, using different data sources, Swamy et al. (2001) study the hypothesis that female participation in the government would reduce corruption. Using World Value Surveys, they find that women are less tolerant towards dishonest behaviour and illegal actions than men. Also, using Georgia Enterprise Survey data, they suggest that men are more frequently involved in corruption than women. Using cross-country data, they conclude that corruption is less prevalent where women carry more weight in politics and in the labour force.

However, Alatas, Cameron, Chaudhuri, Erkal and Gangadharan (2009) conducted an experiment to investigate whether there exist gender differences in corruption activities using Australia, India, Indonesia and Singapore. Interestingly, they find gender differences in only

the Australian sample, where women are found to be less tolerant of corruption than Australian men. They conclude that gender difference in corruption is culture specific. This aligns with Turiel (1994), who states “societal and cultural norms exert a direct impact on moral standards by delineating role and moral responsibilities that, in turn, affect behavior” (p.?). According to McCabe, Ingram and Dato-on (2006), biological sex differences do not predict differences in overall ethical perceptions regarding perceived bribery. They assert that when property rights are not well defined in an economy, corruption can grease the firms’ activities regardless of the entrepreneurs’ gender. Gender differences in corruption and credit access relationship may be the result of both biological and social differences: that is, differences in social roles of men and women.

According to Alatas et al. (2009), an individual’s social role and presence in the public domain may play an important part in that individual’s exposure to corruption. They explain that the level of exposure to corruption in daily life may promote a tolerance and an acceptance of corruption that are reflected in norms of behaviour. The reasons for corruption are always contextual, established in a nation's arrangements, bureaucratic conventions, political advancement, and social history ("Helping Countries Combat Corruption: The Role of the World Bank," 1997). Hence, we develop our second hypothesis as follows:

H₂: There are significant gender differences in paying bribes and obtaining access to credit in SMEs in South Asia.

Data

Data are drawn from the World Bank 2014 Enterprise Surveys⁵, which collect data from micro, small, medium and large firms. The Enterprise Surveys include both qualitative and

⁵ The World Bank’s Enterprise Surveys offer an expansive array of economic data on 130,000 firms in 135 countries. The World Bank Enterprise Survey website provides details as to how the surveys are conducted. (<http://www.enterprisesurveys.org>). An Enterprise Survey is a firm-level survey of a representative sample

quantitative information at the firm level. These surveys comprise representative random samples of firms with data collected across the world, using the same core questionnaire and the same sampling method. The Enterprise Surveys incorporate interviews with business owners and top managers for South Asian countries, of which we examine five: Afghanistan, Bangladesh, India, Nepal, and Pakistan.

<< ***Insert Table1: Total study sample in here***>>

The net result is a sample of 9780 firm level observations. However, only 9667 firm level observation are used for our analysis due to missing observations. From that, 1392 firms have at least one female owner/manager and 8275 SMEs are owned by males (no female owner/manager).

Dependent variable

The dependent variable is *Constraints*, which captures constraints when a firm: (i) applies for and is denied credit (applicants), or (ii) does not apply for credit due to application procedures being complex, collateral requirements being too high/does not have guarantor or size of loan and maturity being insufficient (non-applicant). This taxonomy, consistent with Hewa-Wellalage and Locke (2016) and Hansen and Rand (2014), excludes firms responding “interest rates are not favourable,” “do not think the loan will be approved,” or “there is no need for a loan-establishment as the firm has sufficient capital.” Our sample is carefully chosen in order to minimise the probability of selection-based endogeneity⁶. Neglecting selection biases represents a specification error that is akin to omitted-variable bias endogeneity (Antonakis,

of an economy’s private sector. The surveys cover a broad range of business environment topics including access to finance, corruption, infrastructure, crime, competition, and performance measures.

⁶ Selection-based endogeneity manifests in two main forms: sample-selection and self-selection biases. Heckman’s (1976, 1979) foundational work is principally motivated by sample-selection problems, as samples can be non-representative of a true population and thus threaten both internal and external validity—see Berk (1983) for an excellent review of this issue. However, Heckman (1979) is also conscious of the analogous self-selection problem.

Bendahan, Jacquart, & Lalive, 2010). The basic insight behind selection bias, a form of omitted variable bias, is that the selection process represents an excluded variable that manifests in the error term and correlates with the endogenous choice construct and the outcome variable.

Explanatory variables

Our main independent variable (*Bribe_D*) is derived from the following survey question:

It is said that establishments are sometimes required to make gifts or informal payments to public officials to “get things done” with regard to customs, taxes, licenses, regulations, services etc. On average, what of percentage total annual sales, or estimated total annual value, do establishments like this one pay in informal payments or gifts to public officials for this purpose?

In line with the best practice in corruption studies, the question refers to “establishments like this” to help elicit truthful responses (Billon & Gillanders, 2014). From this question we calculate the following variable:

Bribe_D: This variable determines paying a bribe or not. This takes value 1 if the firm makes gifts or informal payments to public officials to “get things done” with regard to customs, taxes, licenses, regulations, services etc., otherwise 0.

Due to high political unrest and insecure environment in South Asia, most of the SME owners may decline to disclose exact bribe amounts. This is clearly visible in the female owned SME group, so we measure responses relating to paying bribes (or not) as a dummy variable.

Table 2 provides definitions and metrics for variables included in the models.

<<Insert Table 2: Definition and measurement of variables included in the models here>>

Model

Given the dichotomous nature of the credit constraints, a qualitative response model is appropriate. Qualitative response models relate the probability of an event for various independent variables. Such models are often useful when assessing firm characteristics associated with credit access decisions. We apply a discrete choice probit model for binary

choice (yes, no) responses to the credit constraints question. In the binary probit model, credit constraint is shown as 1, while access to credit is 0. The Probit model is written as follows.

$$p(Y=1| X_1, X_2, \dots, X_k) = \Phi(z) = \Phi(\beta_0 + \beta_1.X_1 + \beta_2.X_2 + \dots + \beta_k.X_k)$$

where, p is the outcome of the dummy (0-1) variable for the k^{th} observation, Φ is the standard cumulative normal, X_k is the vector of explanatory variables for observation k and β is the vector of coefficients to be estimated. The Probit coefficients are not directly interpretable, but marginal effects for continuous variables can be calculated (at the mean) as:

$$\frac{\partial \Phi(x_\beta)}{\partial x_k} \Big| = \phi(\bar{x} \beta) \beta_k$$

where, X_k is a vector of independent variables (k is the number of independent variables), β is the vector of estimated coefficients, and ϕ is the normal density function. For dummy variables, the discrete change in probability when the dummy variable switches from zero to one is calculated as $\Phi(X_1\beta) - \Phi(\bar{x})$ where, $X_1 = X_0 = X_k$ except that the i^{th} elements of X_1 and X_0 are set to 1 and 0 respectively (StataCorp, 1997).

Empirical findings related to corruption and credit constraints are mixed. This may be due to endogeneity and reverse causality. Although we carefully selected the sample to minimise endogeneity risk, using the Blundell-Smith (1986) test⁷ we also test endogeneity between the credit constraint proxy and *Bribe_D* variable. The purpose of this test is to provide a simple test of weak exogeneity in limited information simultaneous with limited dependent variable models. The test involves specifying that the exogeneity of one or more explanatory variables is under suspicion. Under the null hypothesis, the models are appropriately specified with all explanatory variables as exogenous. In order to avoid the pernicious effects of weak

⁷ STATA command probexog provides opportunity to test endogeneity in limited dependent variables model (Probit in here).

instruments⁸, the explanatory power and the exclusion restrictions of an instrument are tested before applying the Blundell-Smith test. The test rejects the null hypothesis that the *Bribe_D* variable for both female and male owned SME samples are exogenous at a significance level of 1%. Hence, this study needs to identify possible instrumental variables.

The two requirements for instrumental variables to be valid are (R1) uncorrelated with error term (μ) but (R2) partially and sufficiently strongly correlated with X , once the other independent variables are controlled. The first requirement is difficult to confirm, because we cannot observe μ . So, we rely on economic theory or natural experiments to find instrumental variables that satisfy the first requirement (Muto & Yamano, 2009). Determining whether the selected instruments are relevant in the first stage equation is guided by both a rough “rule-of-thumb” that’s the transformed F -test for the joint significance of identifying instruments in the first stage regression that exceed the value 10, and by reference to a set of critical values provided by Stock and Yogo (2005).

Several macro-level factors may determine the likelihood of paying bribes in SMEs’ ownership and top managerial positions. Corruption will have diverse impacts in different institutional settings and the impacts of corruption on the economy will consequently vary from country to country and from time to time. Concentrating on corruption without taking regard of the interdependencies between corruption and institutions, is improper and may in addition lead to wrong conclusions (Vaal & Ebben, 2011). Hence, this study has identified two potential instrumental variables, which are major representations of the quality of institutional environment of the country.

⁸ The smaller the correlation between instrument and the endogenous variable, the larger the standard errors of the instrumental variable estimator will be. Furthermore, low correlation between the instrument and the endogenous variable can drive to asymptotic biased estimators (Wooldridge, 2002).

This study identifies two potential instrument variables—judiciary system fairness and senior management time. An effective judiciary guarantees fairness in legal processes. It is a powerful weapon against corruption. Judicial system fairness and how it could affect business (*Judiciary*) is an instrument for *Bribe_D*. This variable takes a value of 1 if an SME owner strongly disagrees or tends to disagree with the following statement “*that the court system is fair, impartial and uncorrupted,*” otherwise it takes a value 0. This captures the general perception regarding corruption that firms experience.

In general, the most corrupt nations are those with an extremely weak institutional setting. Senior management’s time spent on dealing with requirements imposed by government regulations (*BG_Relationship*) is an instrument for the variable for *Bribe_D*. *BG_Relationship* variable takes value of 1 if an SME’s senior manager spent more than 50% of his/her time on dealing with government regulations. Evidence exists that SMEs are often denied access to bank loans, which are subject to heavy government regulations (Johnson, Porta, Lopez-de-Silanes, & Shleifer, 2000; McMillan & Woodruff, 2002). If there is complex government regulation (red tape), then the SME owner/manager needs to spend more time dealing with it. Red tape generally includes filling out paperwork, obtaining licenses, having multiple people or committees approve a decision and various low-level rules that make conducting one’s affairs slower, more difficult, or both. If this is the case, a bribe may help to bypass filing and certification requirements, reporting, investigation, inspection and enforcement practices, and procedures.

The *F*-test for instrument relevance for our instruments variable *Judiciary* is 30.11 and *BG_Relationship* is 30.96. Hence both are relevant. Due to lack of firm level observations for the *BG_Relationship* variable, this study continues reporting IV probit estimation with a *Judiciary* instrument variable.

The statistics are far from the rejection of the null, enhancing confidence that the instrument set is appropriate. On the basis of the diagnostic test values, the proposed instruments are well-specified and the econometric findings are robust. It is reasonable to infer that the instrumental variable probit model (i.e., IV probit) is appropriate.

Results

<<*Insert Table3: The share of constrained enterprises and corruption (%) by country here*>>

The figures in Table 3 reveal that corruption and credit constraints are significant in South Asian SMEs. This may be the result of when neighbouring firms take part in corruption and other firms are under pressure to act in similar way. Consequently, corruption is a common practice and has become a necessary instrument for avoiding government red tape. Gender differences in corruption level vary across countries. As approximately 76% of SMEs pay bribes regardless of gender differences in SME owners, this is significantly higher than other South Asia countries. This may illustrate larger countries have relatively more credit constrained firms and the majority need to pay bribes to get things done. Nevertheless, Table 3 shows in Bangladesh, Nepal and Pakistan more female owned SMEs pay bribes to get thing done compared to their male owned counterparts. This may be due to weak institutional structure nepotism in government bodies' lending with preference to male SME owners. When female owned SMEs try to counteract such nepotism, this mostly needs making an informal payment.

<<*Insert Table 4: Descriptive Statistics here*>>

Table 4 provides descriptive statistics for our full SME sample. The variable *Bribe_D* shows approximately 68% of female owned SMEs pay bribes and 69% of male SME owners pay bribes to get things done. This indicates the existence of a high level of corruption, where SMEs make some form of payment to obtain finance and/or pay a regular fee to be and remain in

business in South Asia. Apart from the weak democratic political institutions, cultural factors may have a significant influence in terms of acceptance and practicing bribery as unethical behaviour. According to Transparency International, South Asia is the world's most corrupt region, and South Asia's corruption epidemic is caused by opaque public institutions, lack of protection for anti-corruption actors and widespread government interference in the work of anti-corruption watchdogs ("Fighting Corruption in South Asia: Building Accountability," 2014). Although several studies have shown that women are less prone to corruption (see Agerberg, 2014; Dollar et al., 2001). The figures in Table 4 suggest there is no substantial gender gap in paying bribes in South Asian SMEs. This indicates social and situational factors are likely having a significant impact—that ethicality is influenced by more than biological factors, such as gender. The descriptive characteristics provide additional dimensions for a potentially richer analysis, examining the relationship between credit constraints, corruption and gender in South Asian SME owners.

<<Insert Table 5: Probit estimation results of credit constraints and Corruption relationship in female owned SMEs in South Asia here>>

<<Insert Table 6: Probit estimation results of credit constraints and Corruption relationship in male owned SMEs in South Asia here>>

The probit estimation results for female and male SME groups' credit constraints are reported in Table 5 and Table 6. Column 2 provides probit estimation results measuring bribes as a dummy variable and column 3 provides the marginal effect. The Blundell-Smith test for exogeneity suggests that the unobserved covariates that determined both the corruption and credit constraints are correlated. Therefore, all the results support using IV probit and IV probit marginal effect and these are reported in columns 4 and 5.

Results indicate that *Judiciary* (the instrument of the *Bribe_D*) is negatively correlated with SMEs' credit constraints and is significant at the 1% level for both the male and female group.

Specifically, a corrupted legal system decreases the standardised IV probit index by about 2.121 of an SD in the female group and 1.663 of an SD in the male owned SME group. Further, considering marginal effect, as shown in column 5, when female SME owners pay bribes they have a 71% lower probability of being credit constrained than female SME owners who do not pay bribes. Also, considering the male SME owner group, when they pay bribes they have 58% point lower probability of being credit constrained than their counterparts. This may be associated with a corrupt court system and/or a weak legal system (too weak to secure property rights and enforce contracts). Ideological discrimination against accessing credit by female SME owners disappears when they pay bribes. This finding is consistent with Goetz (2007), who states that corruption is genderless: perhaps corrupt institutions corrupt the people working in them regardless of gender or differences in corrupt behaviour may be more related to nurture and culture than nature.

The risks are shrinking in autocratic states where bribery and favouritism are often a normal part of doing business (Treisman, 2007); indeed, not being corrupt may be riskier than corruption in this context. Women may be differentially impacted by these risks and thus feel greater pressure to conform to existing political norms about corruption. This may be attributable to women being more risk averse when compared to men (Watson & McNaughton, 2007). Women are also more vulnerable to punishment for violating political norms because of explicit or tacit sex discrimination (Stolberg, 2011). As a result, following Esarey and Chirillo (2013), our current study accepts the concept that women are equally susceptible in autocratic systems.

The explanatory variables, firm age (*lnage*) and top manager experience (*lntop_exp*), significantly impact female firms' credit constraints. In male owned SMEs, firm location (*Capital_City*), firm size (*Small*), firm ownership type (*Sole_Pop*) and top manager experience (*lntop_exp*) are significant factors, impacting credit constraints.

Overall, our results reject hypothesis 1 (H_1 : Paying bribes is positively related to SMEs' level of credit constraints). The moral judgement on corruption may bias the understanding of its economic consequences. In a weak institutional environment, corruption may take place in parallel with a low quality of governance and can reduce the inconvenience of such low quality. This aligns with the "grease the wheels" proposition. Further, this study supports the acceptance of hypothesis 2 (H_2 : There are significant gender differences in paying bribes and obtaining access to credit in SMEs in South Asia). This encourages postulating further that corruption may impact differently for male and female SME owners. Surprisingly, this study confirms that when women pay bribes they gain more access to credit compared to their male counterparts.

Conclusion

Gender and corruption linkages are recent issues in the anti-corruption literature. Initially, our research focuses on whether paying bribes is positively related to SMEs' level of credit constraints in South Asia. A second line of enquiry examines whether women are more or less corruptible than men and the impact of corruption on women SME owners, as a group, in accessing credit. Both of these strands of research may lead to generating a wealth of policy-relevant insights that advance our understanding of the interplay between corruption and gender. Micro econometrics modelling, which recognises the likely endogeneity of corruption in credit constraints, and minimises both the endogeneity problem and reverse causality issues, were used via an instrumental variable approach. Support for a "grease the wheels" proposition is observed in the findings that paying bribes increases credit access for SMEs in South Asia. Nevertheless, female entrepreneurs appear to gain more than male entrepreneurs by paying bribes to gain credit for their SMEs. This may be because bribe payments might avoid ideological discrimination for female SME owners, and indicates that bribes become a greater incentive to female SME owners for getting access to credit than their male counterparts.

According to the results, a significant proportion of SMEs pay bribes in South Asia. Therefore, policies focusing on improving the quality of the judiciary, and implementing those policies that are likely to lead to a reduction in corruption levels would be most efficacious. Our research indicates that anti-corruption measures are essential for developing SMEs in South Asia to ensure resources go to stimulate economic growth and achieve the best returns. Women constitute a greater proportion of the population than men in South Asia. In situations where government officials and lending institutions' decision makers are predominantly men, opportunities arise to abuse entrusted power and to expose women SME owners to bribe-paying practises. Introducing a gender focus in anti-corruption measures through mainstreaming gender equality is important for social and economic well-being.

Due to data limitation we could not measure bribe intensity or types of bribes. Future studies may consider bribe intensity and type in the future. There is a need for future research to encompass other continents, using the same methodology employed in this paper, to see whether the “grease the wheel” proposition is reflected beyond South Asia.

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Table 1: Total study sample

Country	Small Firms	Medium size Firms	Total
Afghanistan	256	123	379
Bangladesh	498	515	1013
India	2845	4133	6978
Nepal	283	147	430
Pakistan	509	471	980
	Total number of observations		9780

Table 2: Definition and measurement of variables included in the models

Explanatory variables	Definition	Measurement
<i>Constraints</i>	This variable captures constraints when a firm: (i) applies for and is denied credit (applicants), or (ii) does not apply for credit due to application procedures	1=yes; 0=otherwise

	being complex, collateral requirements are too high/does not have guarantor or size of loan and maturity are insufficient (non-applicant).	
<i>Bribe_D</i>	This variable determines the existence of a bribe.	1=yes; 0=otherwise
<i>Capital_City</i>	Firms from capital city.	1=yes; 0=otherwise
<i>Small</i>	Firm has 5 to 19 employees.	1=yes; 0=otherwise
<i>Medium</i>	Firm has between 20 and 99 employees.	1=yes; 0=otherwise
<i>Sole-Prop</i>	Firm legal status is sole proprietorship.	1=yes; 0=otherwise
<i>lnage</i>	Log number of years since firm has been established	Year(s)
<i>Intop_exp</i>	Log number of years' experience of the firm's manager.	Year(s)
<i>Manufacturing</i>	Firm from manufacturing industry.	1=yes; 0=otherwise
<u>Instrument variables</u>		
<i>Judiciary</i>	The level of judiciary system corruption, based on the answer for following question: Does SME owner strongly disagrees or tends to disagree that the court system is fair, impartial and uncorrupted	1=yes; 0=otherwise
<i>BG_Relationship</i>	Percentage of senior management's time was spent on dealing with requirements imposed by government regulations. This is converted as a Dichotomous variable as follows: If percentage of senior management's time was spent on dealing with requirements imposed by government regulations is greater than 50% or not	1= yes 0= otherwise

Table 3: The share of constrained enterprises and corruption (%) by country

Country	Corruption level		Credit constraints	
	Male	Female	Male	Female
<i>Afghanistan</i>	.4954	.4393	.2826	.2727
<i>Bangladesh</i>	.5138	.6103	.2297	.1818
<i>India</i>	.7669	.7536	.4107	.4059

<i>Nepal</i>	.1372	.6212	.1527	.1666
<i>Pakistan</i>	.3635	.4703	.1371	.3321

Table 4: Descriptive Statistics

Variables	Female					Male				
	Obs	Mean	Std.dev	Min	Max	Obs	Mean	Std.dev	Min	Max
<i>Constraints</i>	1421	.1991	.3995	0	1	8431	.2098	.4072	0	1
<i>Bribe_D</i>	1421	.6812	.4661	0	1	8431	.6949	.4604	0	1
<i>Capital_City</i>	1421	.1309	.3374	0	1	8431	.1475	.3547	0	1
<i>Small</i>	1421	.4588	.4984	0	1	8431	.4846	.4997	0	1
<i>Medium</i>	1421	.5412	.4985	0	1	8431	.5154	.4998	0	1
<i>Sole-Prop</i>	1421	.5714	.4950	0	1	8431	.5849	.4928	0	1
<i>Firm_age</i>	1400	20.5	13.36	2	100	8341	20.16	13.30	2	142
<i>lnage</i>	1400	2.775	.6909	.6931	4.605	8431	2.795	.6672	.6931	4.956
<i>Manager_exp</i>	1413	14.02	9.375	1	64	8354	14.18	9.647	1	65
<i>Intop_exp</i>	1413	2.389	.7714	0	4.158	8354	2.405	.7539	0	4.174
<i>Manufacturing</i>	1421	.7544	.4306	0	1	8431	.7552	.4300	0	1
<i>Afghanistan</i>	1421	.0464	.2105	0	1	8431	.0390	.1936	0	1
<i>Bangladesh</i>	1421	.1084	.3110	0	1	8431	.1068	.3089	0	1
<i>India</i>	1421	.7142	.4519	0	1	8431	.7265	.4459	0	1
<i>Nepal</i>	1421	1421	.0464	0	1	8431	.0411	.1987	0	1
<i>Pakistan</i>	1421	.0844	.2781	0	1	8431	.0864	.2811	0	1

Table 5: Probit estimation results of credit constraints and corruption relationship in female owned SMEs in South Asia

Variables	Probit	Marginal	IV Probit	IV marginal Probit
(I)	(II)	(III)	(IV)	(V)
Bribe_D	.0391 (.0875)			
<i>Judiciary</i>			-2.121*** (.5789)	-.7070*** (.0879)
Capital_City	-.2452* (.1356)	-.0677* (.0373)	-.1547 (.1811)	-.0583 (.0570)
Small	.1577* (.0811)	.0435* (.0223)	.0871 (.1191)	.0334 (.0410)
Sole-Prop	-.0250 (.0819)	-.0069 (.0226)	.1559* (.0848)	.0596 (.0403)
lnage	-.1068 (.1473)	-.0295 (.0406)	-.2053* (.1130)	-.0788* (.0416)
Intop_exp	.2294* (.1264)	.0633* (.0348)	.2993* (.1372)	.1149*** (.0399)
Manufacturing	.1366 (.0953)	.0377 (.0262)	-.0497 (.1406)	-.0191 (.0573)
Afghanistan	.3236 (.2547)	.0893 (.0702)	-.3857 (.4046)	-.1382 (.1664)
Bangladesh	.1553 (.2258)	.0428 (.0623)	.0534 (.1941)	.0206 (.0735)
India	.0885 (.1993)	.0244 (.0550)	.3481** (.1429)	.1297** (.0614)
Pakistan	-.1692 (.2540)	-.0467 (.0701)	-.5341*** (.1774)	-.1858** (.0784)
Summary statistics				
Pseudo-R square	.0155			
Log likelihood	-686.6		-1529.5	
Sample size	1392	1392	1392	
Wald test of exogeneity			7.60***	

Note: Probit regressions. The dependent variable is a dummy variable taking 1 if the SME faces credit constraints and zero otherwise. Column II reports first stage probit regression results, column III reports marginal effects, column IV reports IV probit regression results using *Judiciary* as an instrumental variable. Column V reports IV probit marginal regression results. These models provide standard errors, which are in parentheses. The Wald test of exogeneity is reported in the last row as a chi-squared statistic with 1 degree of freedom. * Significant at 10% level, **Significant at 5% level, ***Significant at 1% level.

Table 6: Probit estimation results of credit constraints and corruption relationship in male owned SMEs in South Asia

Variables	Probit	Marginal	IV Probit	IV marginal Probit
Bribe_D	-.0428 (.0357)	-.0122 (.0102)		
<i>Judiciary</i>			-1.663*** (.3427)	-.5812*** (.1171)
Capital_City	-.2911*** (.0538)	-.0830*** (.0153)	-.2022*** (.0642)	-.0654*** (.0165)
Small	.1111*** (.0326)	.0317*** (.0092)	.1647*** (.0281)	.0558*** (.0105)
Sole-Prop	.0859*** (.0334)	.0245** (.0095)	.1886*** (.0322)	.0632*** (.0133)
lnage	.0270 (.0629)	.0077 (.0179)	-.0304 (.0560)	-.0103 (.0192)
Intop_exp	.0995** (.0501)	.0284* (.0142)	.2099*** (.0453)	.0710*** (.0176)
Manufacturing	.1942*** (.0395)	.0553*** (.01123)	.0236 (.0636)	.0079 (.0208)
Afghanistan	.4112*** (.1136)	.1172*** (.0323)	-.2268 (.1984)	-.07180 (.0635)
Bangladesh	.2791*** (.0976)	.0795*** (.0278)	-.2247 (.1560)	-.0719 (.0520)
India	.1147 (.0881)	.0327 (.0251)	.1182 (.0750)	.0392 (.0243)
Pakistan	-.1989** (.1076)	-.0567* (.0307)	-.7459*** (.1339)	-.1991*** (.0436)
Summary statistics				
Pseudo-R square	.0143			
Log likelihood	-420.36		-904.29	
Sample size	8275	8275	8275	
Wald test of exogeneity			8.45***	

Note: Probit regressions. The dependent variable is a dummy variable taking 1 if the SME faces credit constraints and zero otherwise. Column II reports first stage probit regression results, column III reports marginal effects, column IV reports IV probit regression results using *Judiciary* as an instrumental variable. Column V reports IV probit marginal regression results. These models provide standard errors, which are in parentheses. The Wald test of exogeneity is reported in the last row as a chi-squared statistic with 1 degree of freedom. * Significant at 10% level, **Significant at 5% level, ***Significant at 1% level.

